

- 2 (a) In the fruit fly, *Drosophila melanogaster*, eye colour and wing shape are controlled by genes.
- **E/e** are alleles of a gene involved in eye colour.
 - **E** results in red eyes, **e** results in purple eyes.
 - **N/n** are alleles of a gene involved in wing shape.
 - **N** results in normal (functional) wings, **n** results in vestigial (short, non-functional) wings.

These genes are **both** autosomal.

A dihybrid cross was carried out between a fly with red eyes and normal wings and a fly with purple eyes and vestigial wings. Both parents were homozygous for both genes. The offspring from the F1 generation were crossed to obtain the F2 offspring.

The results are shown in Table 2.1.

Table 2.1

F2 offspring phenotype	expected number of individuals	expected F2 ratio	observed number of individuals	observed F2 ratio
red eye, normal wing	225		287	22.1
red eye, vestigial wing		3	13	1.0
purple eye, normal wing	75		17	1.3
purple eye, vestigial wing		1	83	6.4

- (i) Complete the missing expected number of individuals and the expected F2 ratio in Table 2.1.

- (ii) A chi-squared test showed that the results for the F₂ generation in Table 2.1 were significantly different from those expected.

To investigate this, test crosses were carried out using flies taken from the **F₁ generation** and flies that were homozygous recessive for both genes.

The investigator assumed that the genes were **unlinked** and expected a ratio of 1:1:1:1.

Draw a genetic diagram of this test cross. Use the symbols **E/e** and **N/n**.

(b) The results of the test crosses described in (a)(ii) are shown in Table 2.2.

Table 2.2

offspring phenotype	number of individuals
red eye, normal wing	196
red eye, vestigial wing	22
purple eye, normal wing	23
purple eye, vestigial wing	175

- (i) Flies with red eye, vestigial wing and flies with purple eye, normal wing phenotypes are described as recombinant.

Name the stage of meiosis when these recombinants are produced **and** state how this occurs.

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- (ii) Explain why the results in Table 2.2 are different from the expected ratio of 1:1:1:1.

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[Total: 10]